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- PROJECT TITLE: PFSense Firewall Configuration
 - SUBJECT: Cyber Security
 - ASSIGNED BY: Prof. Waseem Ahmed
 - ASSIGNED TO: Rabiyya Arshad
 - ROLL NO: BSIT51F21S053

Introduction:

What is this Project About?

- Configuring and using PFSense firewall for secure network management.
- Demonstrates how to isolate and control network traffic using virtual machines.
- A step toward understanding real-world network security solutions.



PFSense Firewall Overview:

What is PFSense?

An open-source firewall and router software.

Features:

- Traffic Filtering
- Monitoring and Logging

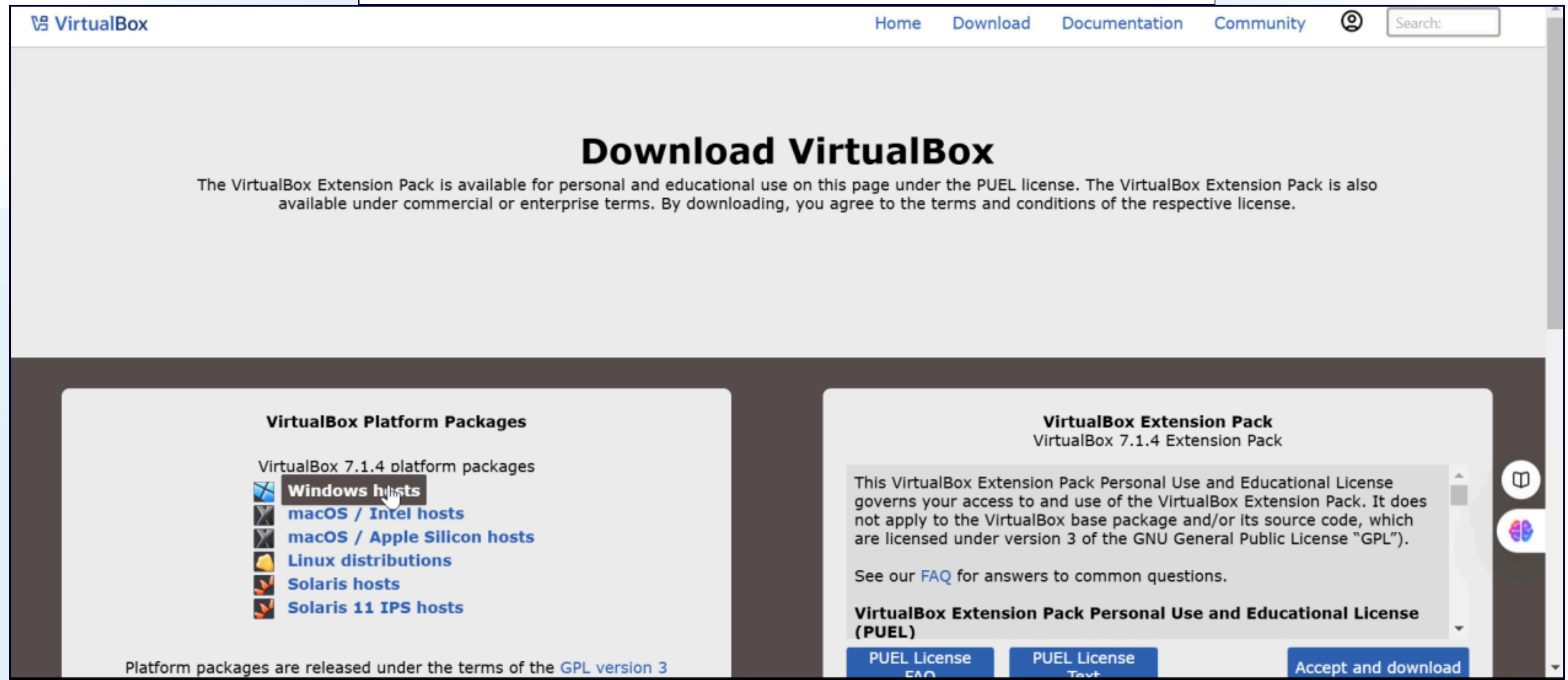


A screenshot of the pfSense Community Edition web interface. The top navigation bar includes links for System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. The main content area is divided into several panels. On the left, there's a 'Dashboard' section showing system information like 'Size' (12G) and 'Usage' (22% of 12G). Below this is a 'System Information' panel showing details like 'Vendor: Phoenix Technologies LTD', 'Version: 6.00', and 'Release Date: Thu Nov 12 2020'. The right side features a 'NTP Status' panel showing 'Server Time' and 'Sync Source'. Below that is an 'Installed Packages' table listing packages like Cron, Open-VM-Tools, and System_Patches. Further down is a 'Services Status' table listing services like dhcpd, dnsmasq, dpinger, ntpd, openvpn, sshd, syslogd, vmware-guestd, and vmware-kmod. At the bottom right, there's a 'Netgate Services And Support' section with a 'Contract type' dropdown set to 'Community Support' and a link to 'NETGATE AND pfSense COMMUNITY SUPPORT RESOURCES'.

Virtual Environment Setup:

Requirements:

- Virtual Box/VMware



The screenshot shows the VirtualBox website's download page. At the top, there's a navigation bar with links for Home, Download, Documentation, and Community, along with a search bar. The main heading is "Download VirtualBox". Below this, a paragraph explains the availability of the VirtualBox Extension Pack under the PUEL license. The page is divided into two main sections: "VirtualBox Platform Packages" and "VirtualBox Extension Pack". The "VirtualBox Platform Packages" section lists various operating systems supported by VirtualBox 7.1.4, with "Windows hosts" highlighted. The "VirtualBox Extension Pack" section provides details about the VirtualBox 7.1.4 Extension Pack, including a link to the PUEL license and a button to "Accept and download".

VirtualBox

Home Download Documentation Community Search:

Download VirtualBox

The VirtualBox Extension Pack is available for personal and educational use on this page under the PUEL license. The VirtualBox Extension Pack is also available under commercial or enterprise terms. By downloading, you agree to the terms and conditions of the respective license.

VirtualBox Platform Packages

VirtualBox 7.1.4 platform packages

-  **Windows hosts**
-  **macOS / Intel hosts**
-  **macOS / Apple Silicon hosts**
-  **Linux distributions**
-  **Solaris hosts**
-  **Solaris 11 IPS hosts**

Platform packages are released under the terms of the [GPL version 3](#)

VirtualBox Extension Pack

VirtualBox 7.1.4 Extension Pack

This VirtualBox Extension Pack Personal Use and Educational License governs your access to and use of the VirtualBox Extension Pack. It does not apply to the VirtualBox base package and/or its source code, which are licensed under version 3 of the GNU General Public License "GPL".

See our [FAQ](#) for answers to common questions.

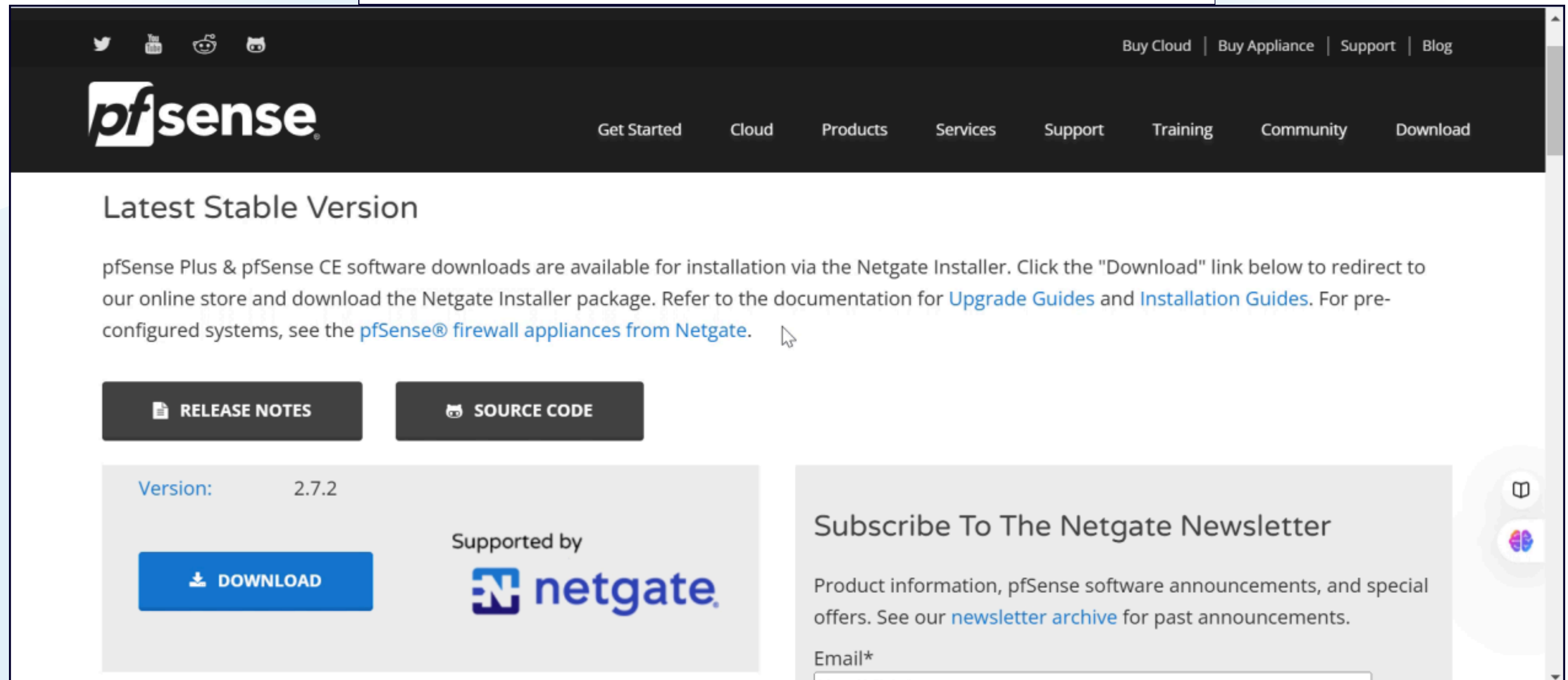
VirtualBox Extension Pack Personal Use and Educational License (PUEL)

[PUEL License FAQ](#) [PUEL License Text](#) [Accept and download](#)

Virtual Environment Setup:

Requirements:

- PFSense ISO Image



The screenshot shows the pfSense website. At the top, there is a dark navigation bar with social media icons (Twitter, YouTube, Reddit, GitHub) on the left and links for 'Buy Cloud', 'Buy Appliance', 'Support', and 'Blog' on the right. Below this is a secondary navigation bar with the pfSense logo and links for 'Get Started', 'Cloud', 'Products', 'Services', 'Support', 'Training', 'Community', and 'Download'. The main content area features the heading 'Latest Stable Version' followed by a paragraph explaining that pfSense Plus & pfSense CE software downloads are available via the Netgate Installer. It includes links for 'Upgrade Guides' and 'Installation Guides', and mentions 'pfSense® firewall appliances from Netgate'. Below the text are two buttons: 'RELEASE NOTES' and 'SOURCE CODE'. A section displays the 'Version: 2.7.2' and a prominent blue 'DOWNLOAD' button. To the right of the download button is the 'Supported by netgate' logo. On the far right, there is a 'Subscribe To The Netgate Newsletter' section with a description of the newsletter content and an email input field.

Buy Cloud | Buy Appliance | Support | Blog

pfsense

Get Started Cloud Products Services Support Training Community Download

Latest Stable Version

pfSense Plus & pfSense CE software downloads are available for installation via the Netgate Installer. Click the "Download" link below to redirect to our online store and download the Netgate Installer package. Refer to the documentation for [Upgrade Guides](#) and [Installation Guides](#). For pre-configured systems, see the [pfSense® firewall appliances from Netgate](#).

 **RELEASE NOTES**

 **SOURCE CODE**

Version: 2.7.2

 **DOWNLOAD**

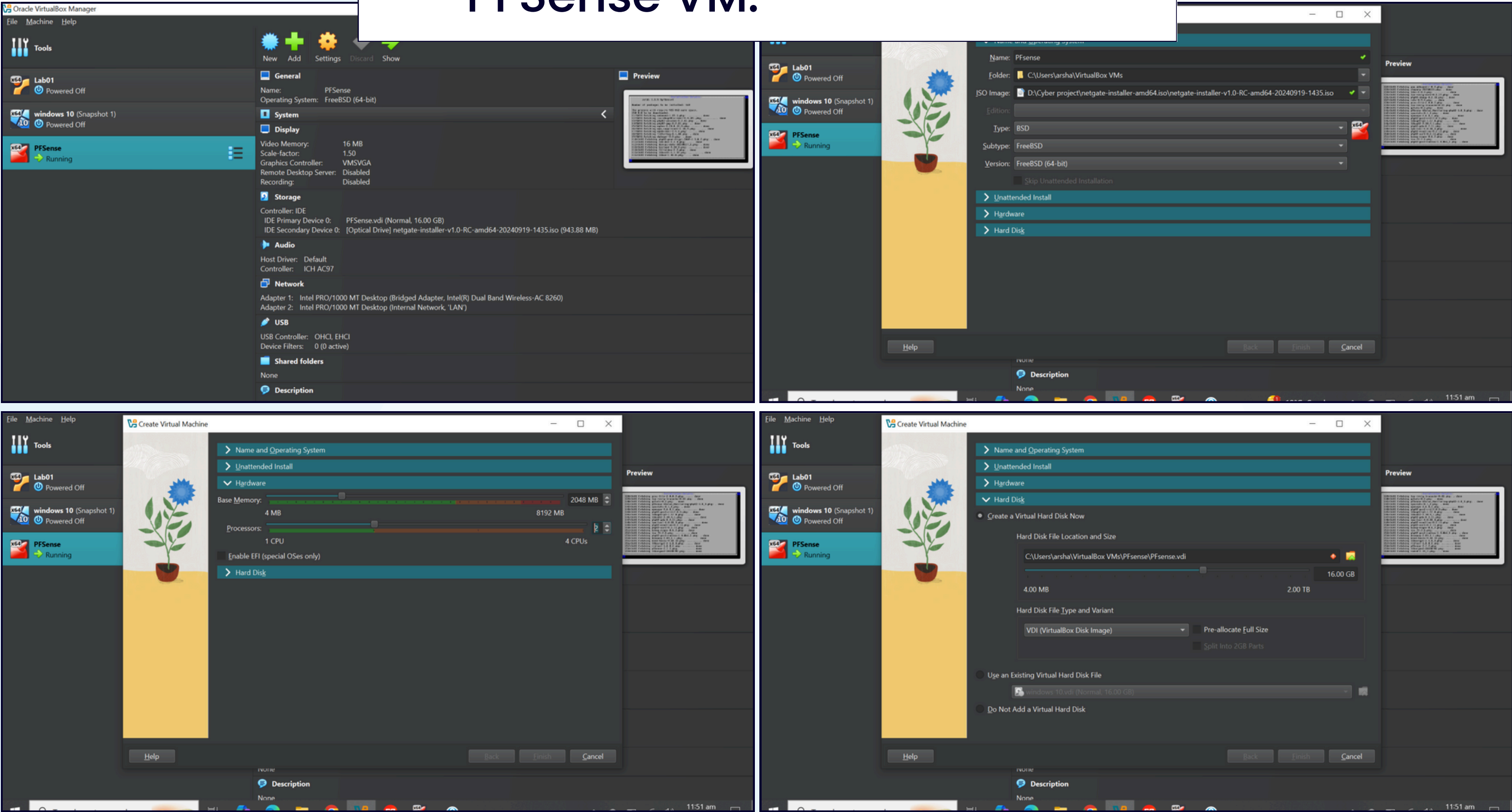
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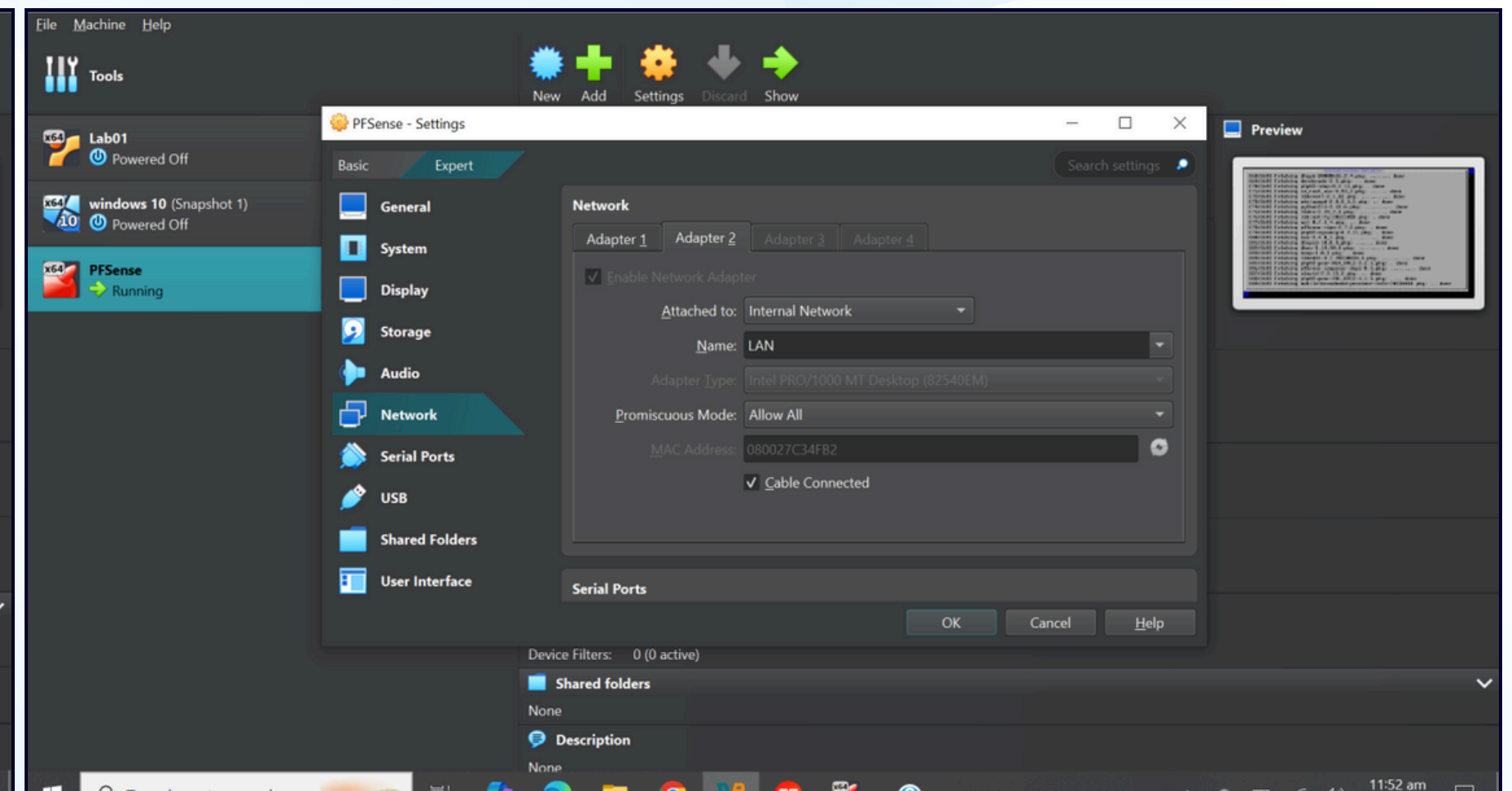
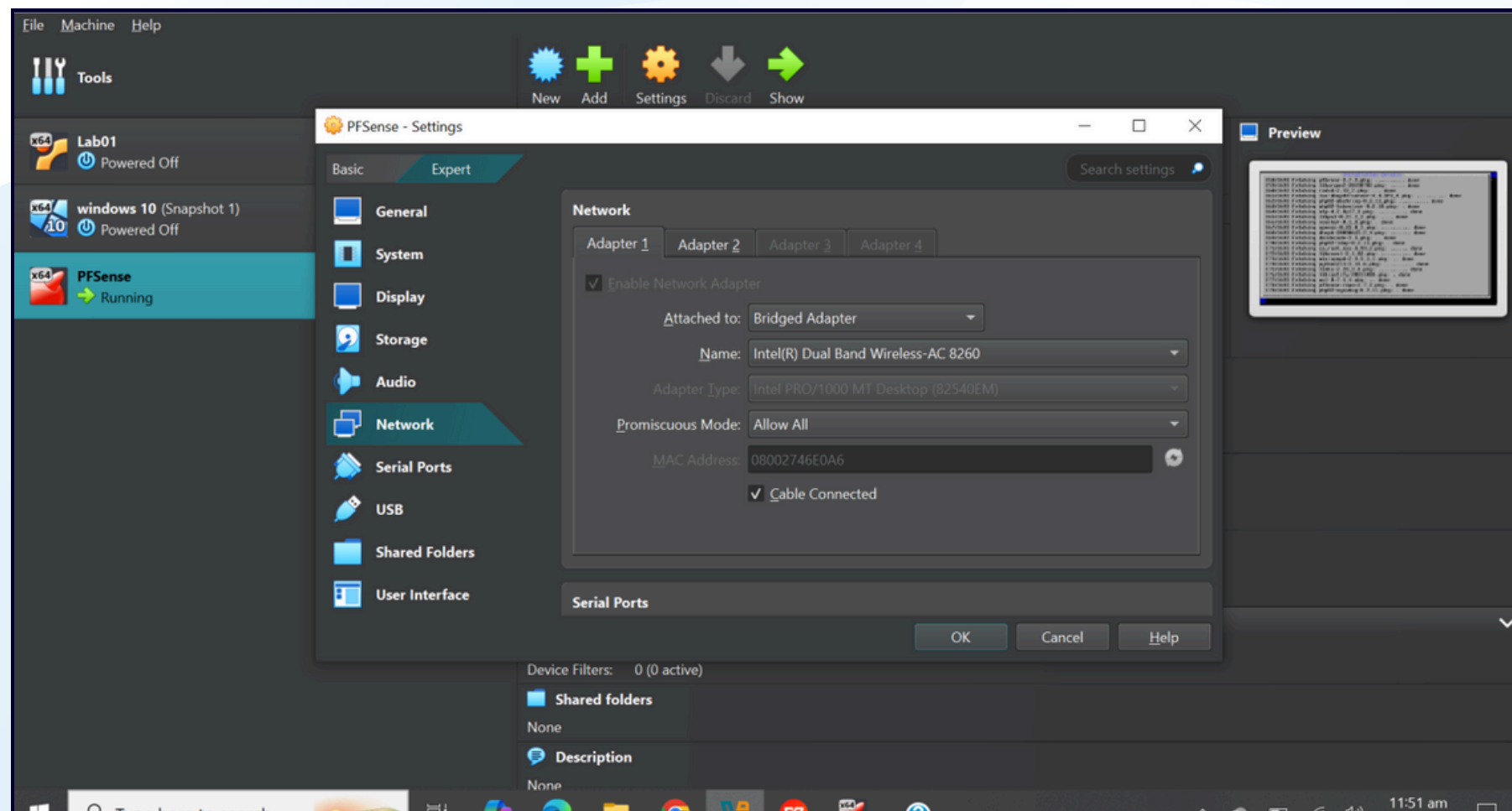
Email*

Components in VirtualBox: PFSense VM:



Network Adapters:

- Interface 1: WAN (Bridged Network).
- Interface 2: LAN (Internal Network).



Windows Host VM

File Machine Help

Tools

Lab01
Powered Off

windows 10 (Snapshot 1)
Running

PFSense
Powered Off

New Add Settings Discard Show

General

Name: windows 10
Operating System: Windows 10 (64-bit)

System

Display

Video Memory: 128 MB
Graphics Controller: VBoxSVGA
Remote Desktop Server: Disabled
Recording: Disabled

Storage

Controller: SATA
SATA Port 0: windows 10.vdi (Normal, 16.00 GB)
SATA Port 1: [Optical Drive] VBoxGuestAdditions.iso (56.11 MB)

Audio

Host Driver: Default
Controller: Intel HD Audio

Network

Adapter 1: Intel PRO/1000 MT Desktop (Internal Network, 'LAN')
Adapter 2: Intel PRO/1000 MT Desktop (NAT)

USB

USB Controller: xHCI
Device Filters: 0 (0 active)

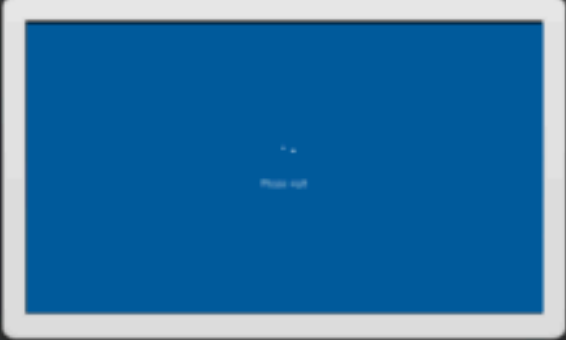
Shared folders

None

Description

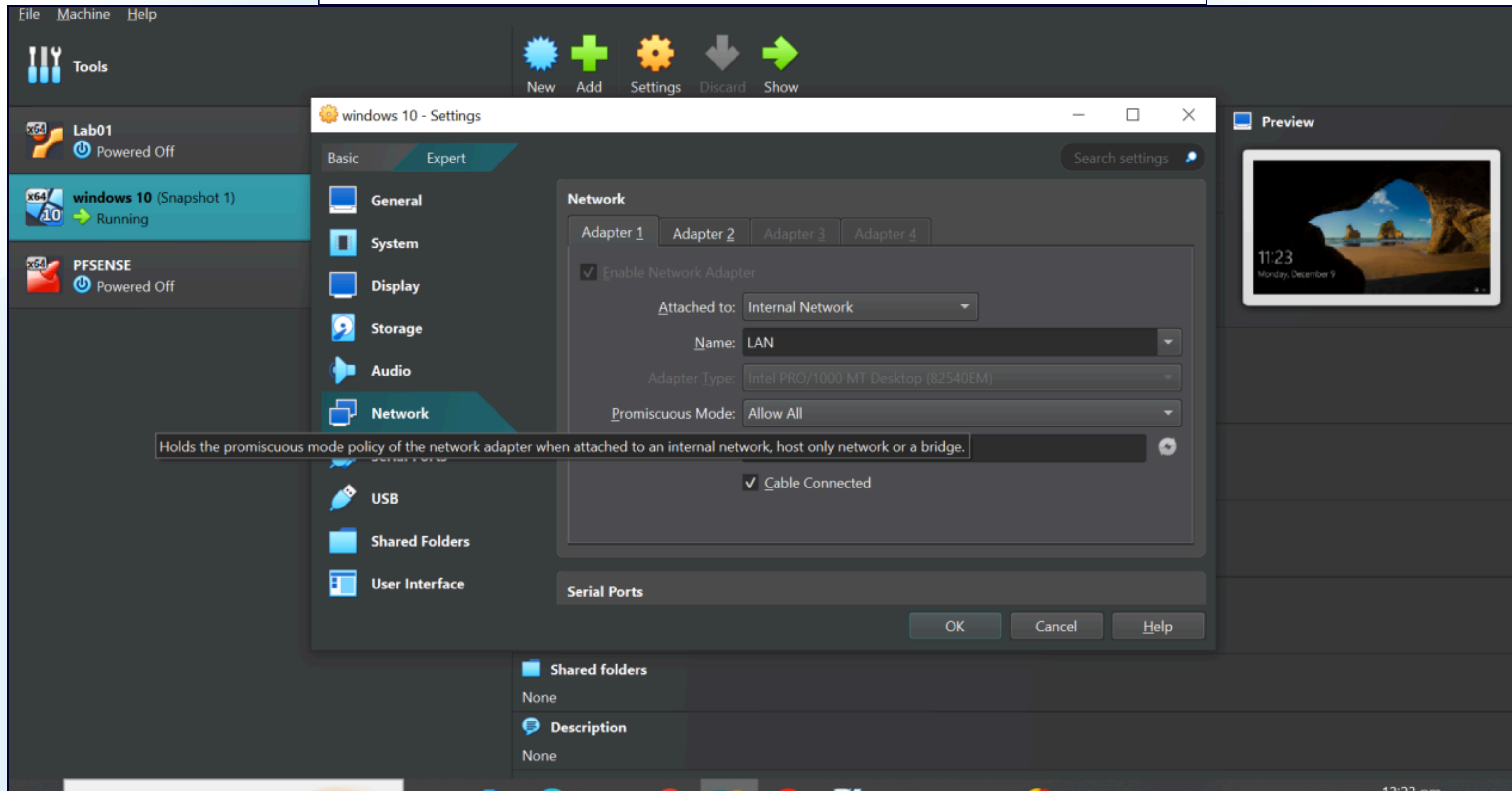
None

Preview



Network Adapters:

- Connected to LAN Internal Network.



After all the settings of network adapters click to start the machine and by continue with default settings setup pfsense.

PFSense VM's Network Configuration:

WAN Interface (PFSense):

- IP Address: 192.168.157.153/24.

```
Enter an option: 2
```

```
Available interfaces:
```

```
1 - WAN (em0 - static)
```

```
2 - LAN (em1 - static)
```

```
Enter the number of the interface you wish to configure: 1
```

```
Configure IPv4 address WAN interface via DHCP? (y/n) n
```

```
Enter the new WAN IPv4 address. Press <ENTER> for none:
```

```
> 192.168.157.153/24
```

```
For a WAN, enter the new WAN IPv4 upstream gateway address.
```

```
For a LAN, press <ENTER> for none:
```

```
>
```

```
Configure IPv6 address WAN interface via DHCP6? (y/n) n
```

```
Enter the new WAN IPv6 address. Press <ENTER> for none:
```

```
>
```

```
Do you want to enable the DHCP server on WAN? (y/n) n
```

PFSense VM's Network Configuration:

LAN Interface (PFSense):

- IP Address: 172.16.0.1/16.

```
Enter an option: 2
```

```
Available interfaces:
```

```
1 - WAN (em0 - static)
```

```
2 - LAN (em1 - static)
```

```
Enter the number of the interface you wish to configure: 2
```

```
Configure IPv4 address LAN interface via DHCP? (y/n) n
```

```
Enter the new LAN IPv4 address. Press <ENTER> for none:
```

```
> 172.16.0.1/16
```

```
For a WAN, enter the new LAN IPv4 upstream gateway address.
```

```
For a LAN, press <ENTER> for none:
```

```
>
```

```
Configure IPv6 address LAN interface via DHCP6? (y/n) n
```

```
Enter the new LAN IPv6 address. Press <ENTER> for none:
```

```
>
```

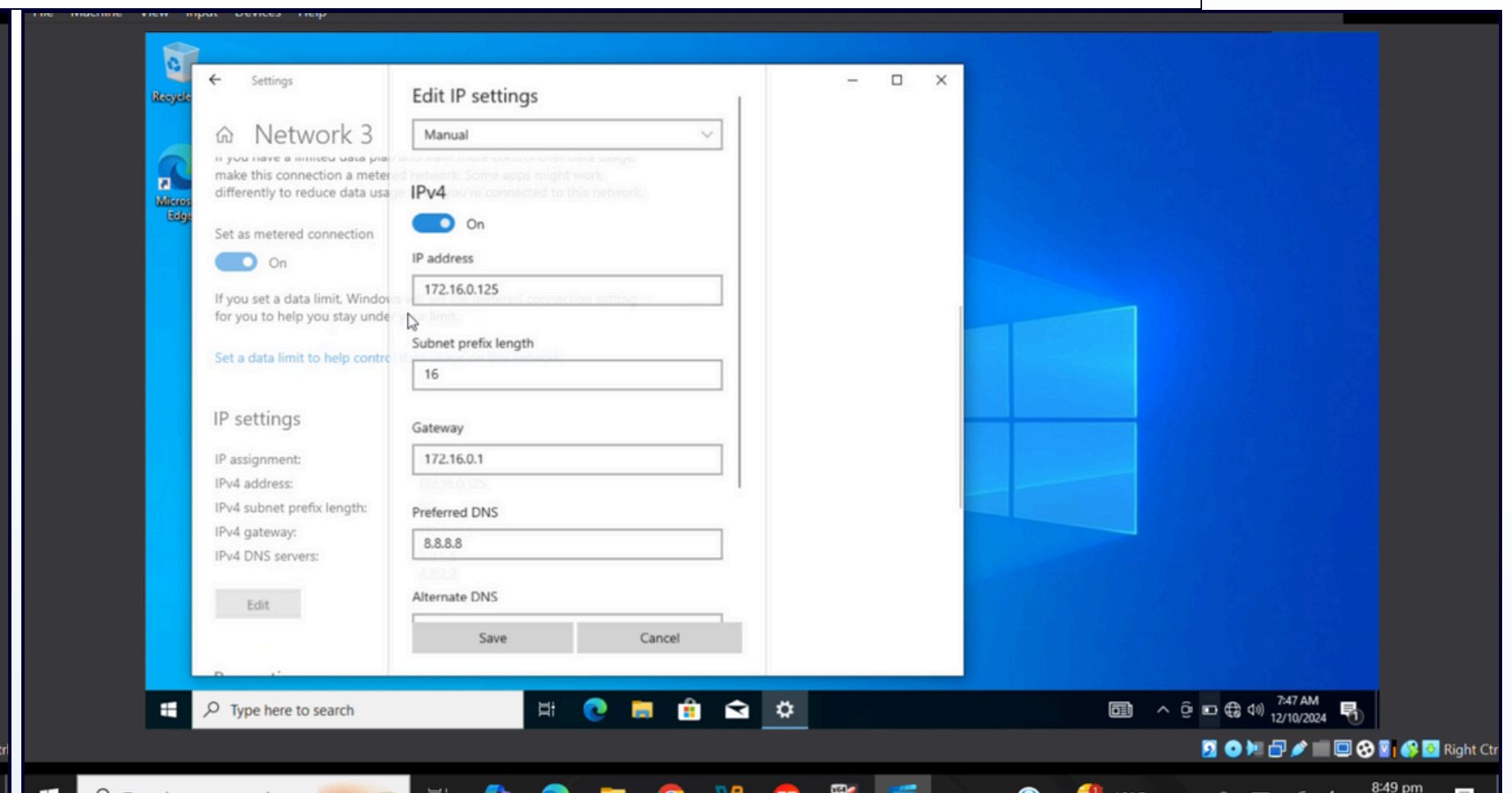
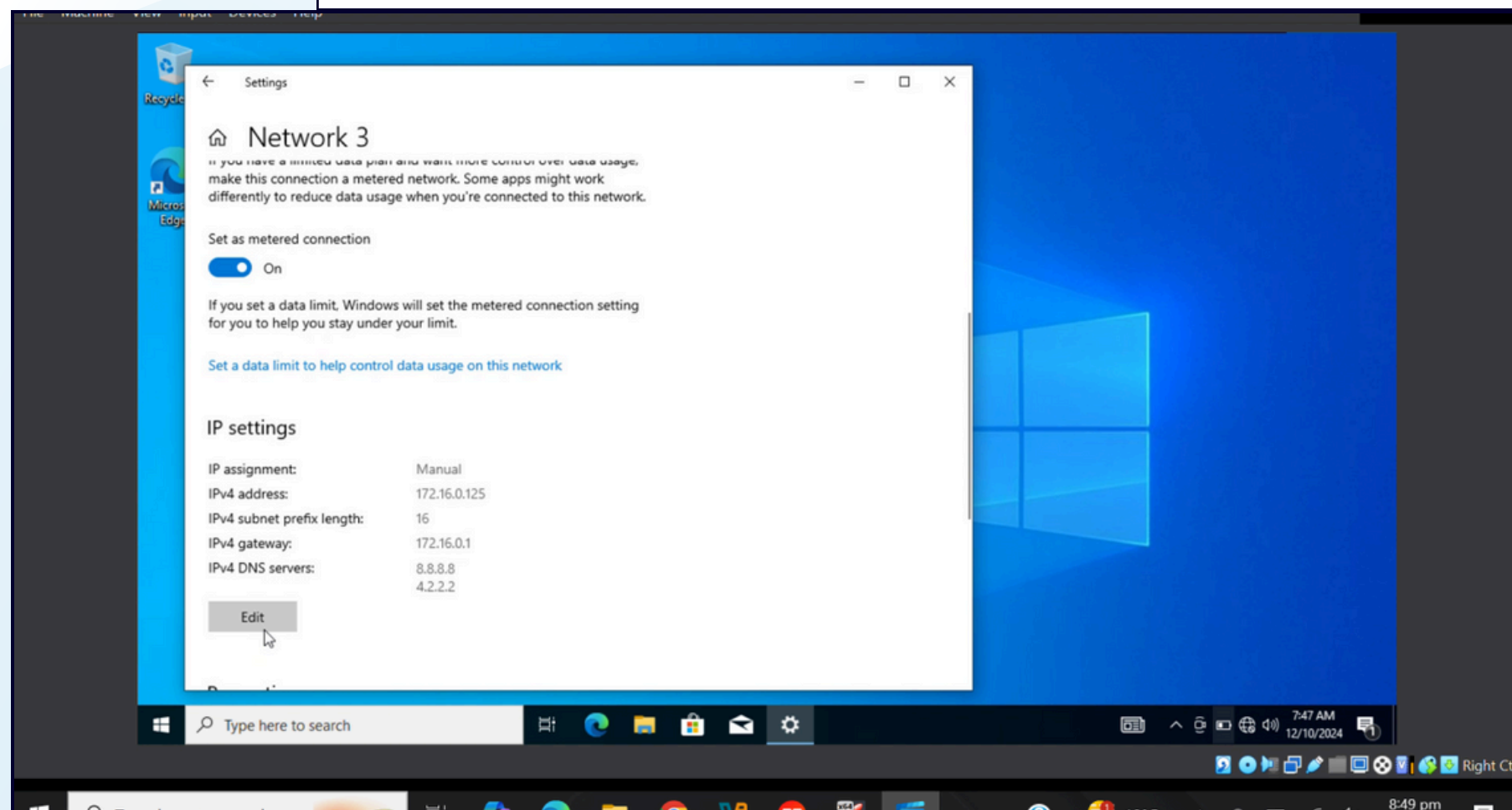
```
Do you want to enable the DHCP server on LAN? (y/n) █
```

Windows Host VM's Network Configuration:

Steps:

Configure:

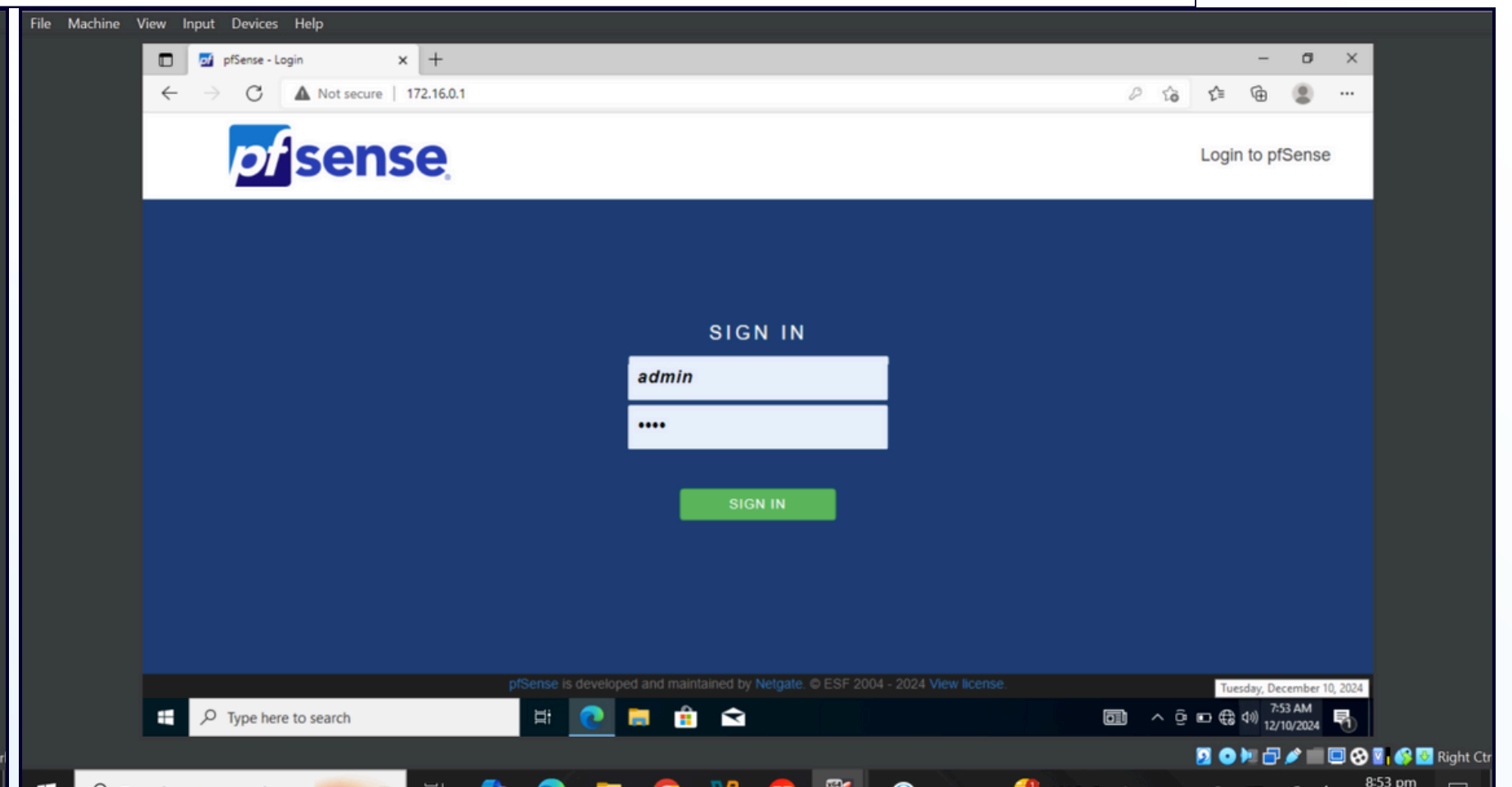
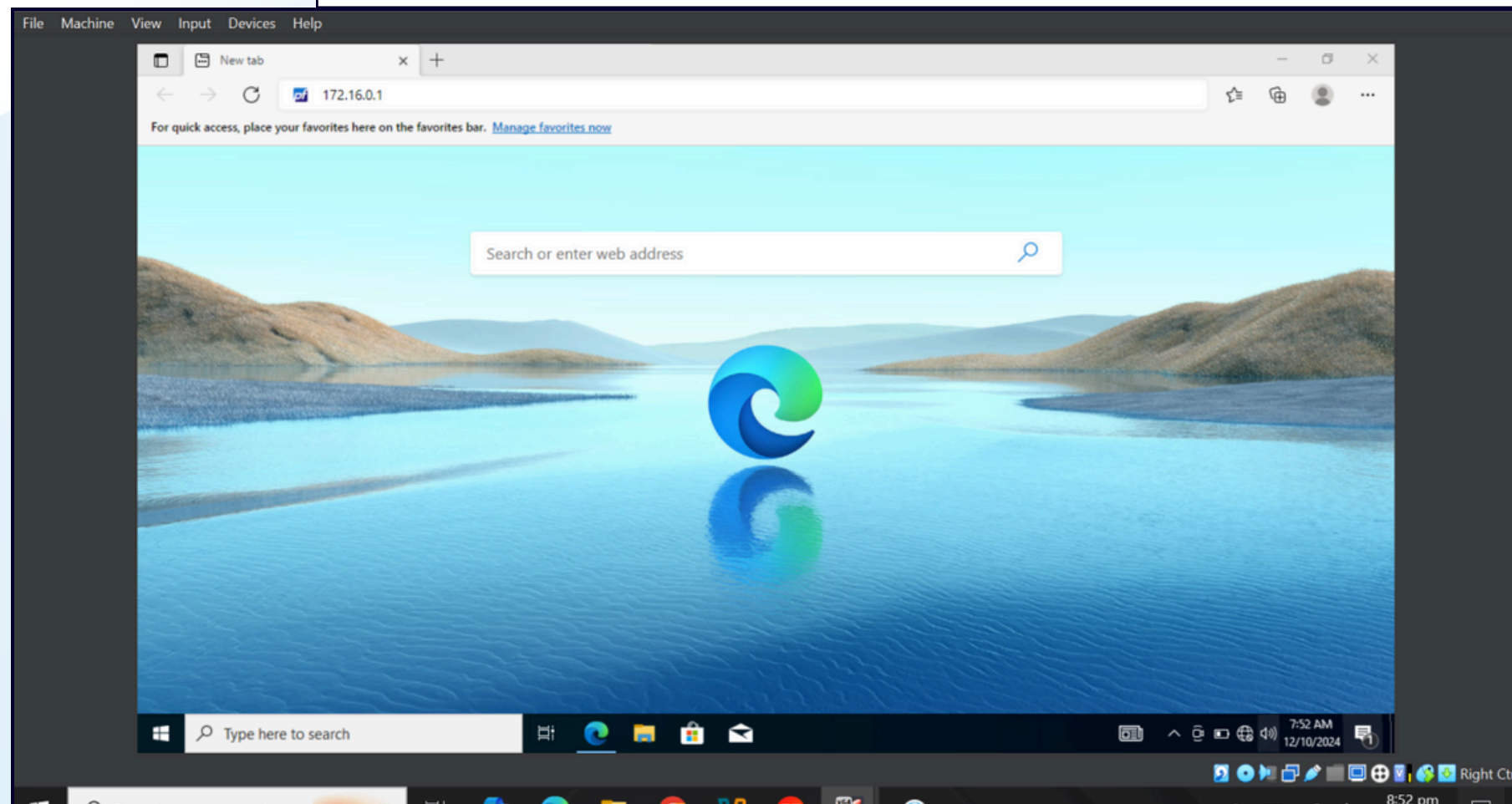
- IP Address: 172.16.0.125
- Subnet Mask: 255.255.0.0
- Default Gateway: 172.16.0.1



Accessing PFSense GUI:

Steps to Access:

- Open a browser on Windows Host VM.
- Enter the LAN IP: 172.16.0.1.
- Login to PFSense with default credentials (admin/pfsense).



Status / Dashboard



System Information



Name	pfSense.home.arpa
User	admin@172.16.0.125 (Local Database)
System	VirtualBox Virtual Machine Netgate Device ID: 5771a519f7620dd6ab94
BIOS	Vendor: innotek GmbH Version: VirtualBox Release Date: Fri Dec 1 2006
Version	2.7.2-RELEASE (amd64) built on Tue Mar 5 0:53:00 PKT 2024 FreeBSD 14.0-CURRENT Unable to check for updates
CPU Type	Intel(R) Core(TM) i5-6300U CPU @ 2.40GHz 2 CPUs: 1 package(s) x 2 cache groups x 1 c AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive

Netgate Services And Support

Retrieving support information 

Interfaces



 WAN		1000baseT <full-duplex>	192.168.157.153
 LAN		1000baseT <full-duplex>	172.16.0.1

PFSense Firewall Rules:

Rule to block facebook.com

- Firewall>Rules>Add Rule to LAN Interface
- Action: Block
- Protocol: TCP/UDP
- Source: LAN Subnets
- Destination: Any
- Destination Port Range: 80 for HTTP – 443 for HTTPS
- Description: Facebook Blocked
- Save and click to apply changes

Rule by DNS Resolver:

- Click add under Host Override option.
- Hostname: Facebook
- Domain Name: facebook.com
- IP Address: 0.0.0.0
- Description: Facebook Blocked
- Click save and apply changes.

Test the Rules by using pfsense shell or DNS Lookup.

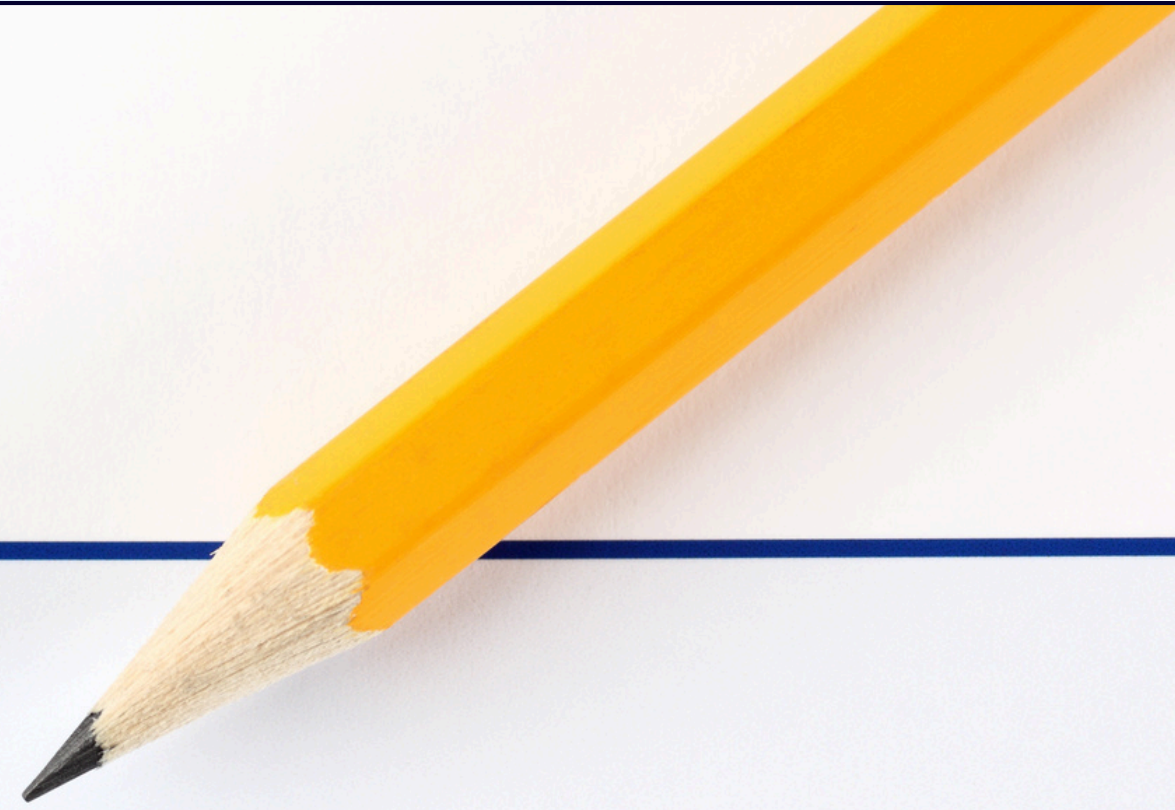
Purpose and Benefits of PFSense:

Why Use PFSense?

- Security: Filters and controls traffic.
- Access Management: Restricts unauthorized users.
- Network Segmentation: Creates isolated networks.
- Monitoring: Logs and monitors traffic for analysis.

Real-Life Use Cases:

- Office network security.
- Educational institution internet management.
- Secure home networks.



THANK YOU